Aditya Rao

647-978-3730 | raoa32@mcmaster.ca | in/aditya-g-rao | adityarao2005 | https://adirao.ca

Education

McMaster University Sep 2023 – Apr 2028 (Expected)
B.Eng in Software Engineering (CO-OP) *Hamilton, ON*

- **3.99/4.00 CGPA** - Dean's Honour List
- **Relevant Coursework:** Data Structures and Algorithms, Concurrent Systems Design, Software Requirements, Large System Design, Computer Architecture, Software Testing, Databases

Experience

IBM May 2025 – Aug 2026
Software Developer Intern, WebSphere Application Server *Markham, ON*

- Rearchitected patch publishing into Java services with Kafka workers, cutting publishing from half a day to ∼10 minutes.
- Built an AI agent diagnosing build failures against logs and source code, saving ∼4 hours per failure.
- Rebuilt customer-issue tracking, preventing synchronization bugs, separating support ticket data from approval requests.
- Added versionless installation to Liberty's installUtility and rewrote WebSphere installation tests in JUnit.
- Replaced spreadsheets with work-allocation tracking for 400+ staff, saving ∼60 minutes per person weekly.
- Delivered 27 merged OpenLiberty PRs and 100+ merged internal PRs across WebSphere and developer tooling.

McMaster University Sep 2026 – Dec 2026
Teaching Assistant - Digital Systems *Hamilton, ON*

- Led digital systems labs for 50+ students, graded prelabs and demos, and supported Verilog design and FPGA debugging.
- Validated and improved lab guides covering embedded C on NIOS II for DRAM read/write testing on DE1-SoC FPGAs.

Medium AI May 2024 – Aug 2024
Full Stack Developer Intern *Hamilton, ON*

- Built real-time doctor–patient transcription using browser audio streaming and load-balanced gRPC Whisper services.
- Implemented FFmpeg audio preprocessing and Firebase synchronization to display live transcript updates.

Projects

BYoAI Deployment Platform | | *Go, TypeScript, Docker, MCP, OpenTelemetry*

- Built an open-source platform deploying AI agents in sandboxed **Docker** environments, using a **Go** daemon (**40 MB RSS**) for isolated execution and a **TypeScript/Bun** service for model orchestration and tool routing.
- Engineered pluggable **event routing** across Redis, RabbitMQ, and Kafka, and a custom **MCP-over-RPC transport** tunneling tool servers inside remote Docker containers.
- Built a React 19 web UI and Go terminal TUI with OAuth2 (confidential + PKCE), declarative tool ACLs, and human-in-the-loop approval gates.

WebCraft | | *C++, CMake, Vcpkg, Asynchronous Programming, Github Actions*

- Authored an asynchronous runtime in **C++23** featuring asynchronous timers, task scheduling, and high-performance file and socket (TCP and UDP) I/O modules.
- Sustained **160K req/s** on synthetic **wrk** HTTP benchmarks via an **io_uring** event loop on Linux.
- Automated CI using **gtest (130+ tests)**, **GitHub Actions CI/CD** (Linux, Windows, macOS).

Resume Builder | | *Java, Spring Boot, Next.js, Typescript, Python, FastAPI, LangChain, Docker*

- Built an AI resume builder with **Next.js**, **Spring Boot**, and **FastAPI** for real-time editing and version control.
- Integrated **LangChain** to automate ATS keyword optimization, intelligent scoring, and template tailoring.

FashioNova | | *Typescript, Next.js, Python, Socket.io, OpenCV, Cloudflare AI, Flask*

- Developed a virtual wardrobe and AI fashion advisor in 36 hours at Hack the Valley 9 at UofT Scarborough
- Created an OpenCV pipeline using Python Flask and Socket.io to fit virtual clothes onto users via live video feed
- Leveraged Cloudflare AI to suggest outfits based on the occasion, promoting sustainable fashion consumption.

Smart Study Buddy | | *Python, Flask, Wolfram Alpha API, OpenAI API, Next.js, Typescript*

- Developed a virtual wardrobe and AI fashion advisor in 36 hours at Hack the Valley 9 at UofT Scarborough
- Created an OpenCV pipeline using Python Flask and Socket.io to fit virtual clothes onto users via live video feed
- Leveraged Cloudflare AI to suggest outfits based on the occasion, promoting sustainable fashion consumption.

QuakeGuard | | *Next.js, Flask, Typescript, Python, TensorFlow, Keras, SciKit Learn, Pandas*

- Developed an earthquake prediction tool using ML models showcased at DeltaHacks X and GDSC McMaster
- Visualized global earthquake data and high-risk areas with Next.js and Python, Flask, SciKit Learn, and TensorFlow

Technical Involvement

Sys-Admin and Fullstack Developer Executive Nov 2024 – Present
IEEE McMaster Student Branch *Hamilton, ON*

- Set up an **RPi 3B+ cluster** handling up to 100 concurrent clients with **<50ms response latency**.
- Provisioned **CI/CD pipelines** to build and deploy container images to a **Kubernetes cluster**.
- Developed the IEEE branch website using Next.js with GitHub Projects/Issues as an admin CMS.

Technical Skills

Languages: Java, JavaScript/TypeScript, Python, C#, C++, Golang, SQL, C, Verilog
Frameworks: Spring Boot, Jakarta EE, MicroProfile, Quarkus, Hibernate, Express, Next.js, FastAPI, Flask
Developer Tools: AWS, GCP, Git, Docker, Kafka, OpenShift, Kubernetes, Redis, PostgreSQL, Infinispan
Libraries: Pandas, TensorFlow, OpenCV, SciKit Learn, Redux, React, Mongoose
Hobbies: Sketching, Painting, Violin (Western and Carnatic), Badminton, Swimming, Chess

Certifications

Clean Code – *Udemy* *September 2025*
IBM Agile Explorer – *IBM* *September 2025*
The DevOps Essentials - The Handbook – *Udemy* *September 2025*
Introduction to Containers, Kubernetes, and OpenShift – *IBM* *August 2025*
Practical OpenShift for Developers - OpenShift 4 – *Udemy* *August 2025*
[NEW] Ultimate AWS Certified Cloud Practitioner CLF-C02 2025 – *Udemy* *February 2025*
Docker & Kubernetes: The Practical Guide [2025 Edition] – *Udemy* *January 2025*